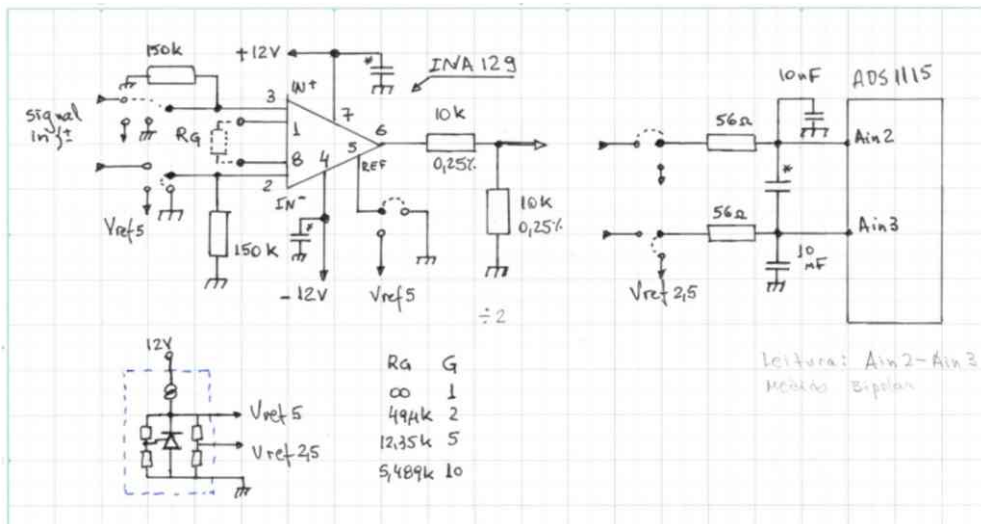


Bipolar Circuit for Monopolar Differential

The circuit presented here translates a signal with an excursion between +5.0 V and -5.0 V referenced to Ground (GND) to a signal between 0 V (Zero Volts) and +10 V, through a "Level Shift" circuit formed by the INA129 instrumentation amplifier (InsAmp).

Problem: Analog-to-digital converters (ADCs) in general are not capable of measuring negative signals relative to Ground (GND); however, many of them allow differential measurements, that is, measuring the signal of one input subtracted from the other. In this differential format, the center of the scale corresponds to the Zero potential. For example, if the measured value is between Zero and 5.0 Volts, the value of 2.5 Volts will correspond to a Zero reading (at the binary output of the ADC). Therefore, values between 2.5 V and zero will correspond to negative values, and values between 2.5 V and 5.0 V will correspond to positive values, using the two's complement binary format. For example, in the ADS115, this data format is described in section 7.5.4 (Data Format) of the respective Data Sheet.

The InsAmp INA129 allows gain adjustment via a single external resistor (R_G) connected between pins 1 and 8. The gain can be adjusted between 1 and 10,000. For unity gain, the input voltage can vary between -5.0 V and +5.0 V. As the gain is increased, the input voltage spacing is reduced proportionally; for example, with a gain of 2, the input can vary between -2.5 V and +2.5 V. The output level shift of the INA129 can be done in two ways: (a) injecting a fixed potential (V_{ref5}) into one of the inputs (pins 2 or 3) or (b) into pin 5 (REF), as adopted in this work.



An offset adjustment is possible via pin 5 (REF) by injecting a small (stable) potential into this pin, similar to that suggested in the Data Sheet (section 8.2.2.3, Offset Trimming).

The input signal can be injected into pin 3 (non-inverting input) and pin 2 (inverting input) connected to ground, and 5.0 V applied to pin 5 (REF). In this way, the output (pin 6) will be the Signal (pin 2) - Zero (pin 2) + V REF (pin 5), i.e., the output voltage will be the input offset by +5.0 V. This output voltage will be divided by 2, since the ADC input is configured for +/- 2.048 V relative to pin Ain3, where the 2.5 V reference voltage is connected. Therefore, the ADC input saturation will occur at +4.5 V on Ain2 for positive saturation (7FFFh) and +0.5 V on Ain2 for negative saturation (8000h). This corresponds to an input signal of +/- 4.0 V.

The ADC input filter (ADS1115) consists of two 56 Ohm resistors and three capacitors (two 10 nF and one 100 nF), all capacitors marked with (*) correspond to 100 nF. This filter is suitable for a sampling rate of 860 SPS (maximum rate of the ADS1115).

References

1. <https://www.ti.com/lit/ds/symlink/ina129.pdf>
2. <https://www.ti.com/lit/ds/symlink/ads1115.pdf>
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